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On the Characterization of Phase Noise for the Robust and Resilient PLL-TRNG Design

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Abstract. A true random number generator (TRNG) is a critical component in ensuring the security of cryptographic systems. Among TRNG implementations, the phase-locked loop-based TRNG (PLL-TRNG) is a widely adopted solution for FPGA platforms due to the availability of a stochastic model. In the previous study, this stochastic model was based on analog noise signals, which potentially led to an oversimplification of the PLL physical process and resulted in an overestimation of entropy. To address this limitation, we extract key platform-specific parameters of the PLL and develop a new stochastic model tailored for multi-output PLL-TRNGs. For the first time, we reveal the effect of the PLL's bandwidth on the correlation of sampling points and introduce a method for quantitatively controlling sampling point correlations. Finally, we validate the model through on-chip jitter measurements. Experimental results show that the proposed stochastic model accurately describes the behavior of the PLL-TRNG and provides the most conservative entropy lower bound, with a 1.8-fold improvement in jitter resolution.

Keywords: True random number generator · Phase noise · Stochastic model · FPGA

1 Introduction

A true random number generator (TRNG) is a fundamental component in cryptographic systems. It extracts randomness from unpredictable physical processes, ensuring that the generated random numbers are non-manipulable. Due to the high quality of the random numbers, TRNGs have a wide range of applications, including key generation, nonce creation, initialization vectors, and generating masks for counteracting side-channel attacks.

Among various TRNG architectures, the phase-locked loop-based TRNG (PLL-TRNG) is a competitive candidate. It utilizes the clock jitter of the phase-locked loop (PLL) as the noise source and performs entropy extraction through coherent sampling. The main advantages of PLL-TRNGs are as follows. The PLL-TRNG offers high security, providing near-maximal entropy density even without the use of post-processing [GGFZ22]. In mainstream FPGA platforms, the PLL and digital logic parts are separately powered and physically isolated, preventing crosstalk and interference between them. Additionally,

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there is no need for manual layout and routing, ensuring reproducibility across different designs [PMB+16]. PLL-TRNGs are characterized by detailed stochastic models and dedicated embedded tests, which enhance their reliability [BBF+18]. Due to these advantages, extensive research has been conducted on the development and analysis of PLL-TRNGs [FD02, FDŠB04, SDFF06, BFV10, FBB19, FBB+23].

According to the design methodology of TRNGs, stochastic models are used to assess their security. This is also a mandatory requirement of the AIS-31 [KS11] standards. In [FBB+23], notable work has been done on the stochastic model to enhance the security of PLL-TRNGs. However, this model fails to quantitatively analyze the dependency of sampling bits and cannot accurately assess the entropy of multiple PLL outputs. Designers cannot estimate the magnitude of bit correlations for a given PLL configuration without relying on hardware implementation. Given that PLLs have a vast configuration space [PMBF17], this adds complexity to the design process. On the other hand, multi-output PLLs inevitably exhibit dependency between different outputs. The assumption of weakly correlated contributors never holds, leading to a significant discrepancy in entropy estimation, as shown in Table 3 of [FBB+23].

The cause of these problems is that the stochastic model proposed in [FBB+23] is based on the assumptions of analog noise signals. These assumptions may overly simplify the analog process of the entropy source, leading to entropy overestimation for TRNGs [HTBF14]. It is also worth noting that previous research assumed the raw random numbers to be independent and identically distributed (IID). Despite the detailed statistical analysis of the raw random numbers in [FBB+23], a more rigorous analysis of the analog processes is still needed to support this assumption. Therefore, a direct analysis of the phase noise process of PLLs is essential for a robust design. Many studies have made significant contributions to the analysis of clock jitter-based entropy sources [VFAB10, BCPW22, PV24a]. However, these works primarily focus on ring oscillators and cannot be directly applied to the more complex structure of PLLs.

In this paper, we characterize the physical platform parameters of the PLL and describe its phase noise process. We also propose an enhanced stochastic model for the PLL-TRNG with multi-outputs. We simulate and analyze the impact of different design and platform parameters on entropy, and validate the results through on-chip jitter measurements. Our contributions are below.

1. We propose an enhanced PLL-TRNG stochastic model, grounded in the characterization of the PLL's key platform-specific parameters. This bridges the gap in the existing lack of stochastic models for multi-output PLL-TRNGs. We share the implementations for the proposed stochastic model and all the necessary material are open source[1].
2. We reveal the effect of the PLL's bandwidth on the correlation of sampling points for the first time and present a quantitative approach for estimating the upper bound of sampling point correlation. This approach improves the sampling point distance control technique in [FBB+23].
3. We put forward an entropy evaluation method incorporating jitter measurements into the stochastic model as shown in Fig. 1. Compared to the latest research, this method achieves approximately a 1.8-fold improvement in jitter resolution and provides the most conservative entropy estimates.

This paper is organized as follows: Section 2 introduces the mathematical foundation and the principle of PLL-TRNGs. Section 3 presents the phase noise process for PLLs and an enhanced stochastic model with multiple PLL outputs is proposed in Section 4. Section 5 focuses on the hardware circuit design of PLL-TRNGs. Finally, Section 6 discusses the validity of the model and the results of the entropy rate evaluation. Section 7

[1] The implementations and material are publicly available: https://github.com/ybhphoenix/PLL-TRNG-Characterization

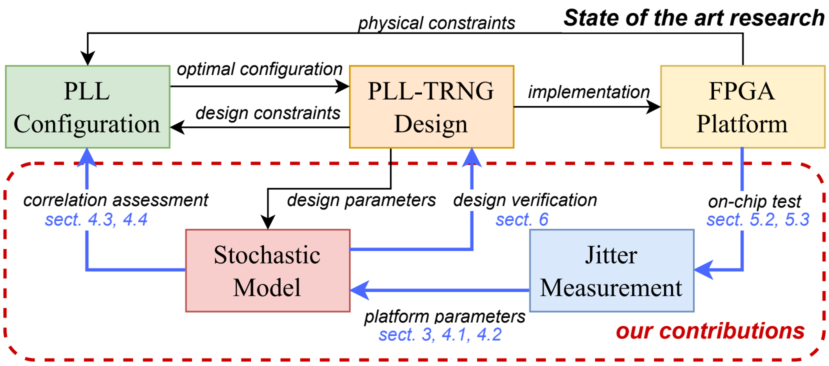


Figure 1: Proposed workflow of the PLL-TRNG design

concludes the paper.

2 Preliminary

2.1 Notion

In this paper, notations and definitions related to random variables and stochastic processes are summarized in Table 1.

Table 1: Notations and Definitions

Notation	Definition
X, Y	Random variables
x, y	Realizations of random variables
$\Pr(\cdot)$	The probability of an event
$f_X(\cdot)$	The probability mass function (PMF) or probability density function (PDF) of X
$\mu_X, \mathbf{E}(X)$	The expected value of X
$\sigma_X^2, \mathbf{Var}(X)$	The variance of X
σ_X	The standard deviation of X
$\mathbf{Cov}(X, Y)$	The covariance between X and Y
ρ_{XY}	The correlation between X and Y
$X(t)$	The stochastic process at time t
$R_X(\tau)$	The autocorrelation function (ACF) of $X(t)$
$S_X(f)$	The power spectral density (PSD) of $X(t)$
$\mathcal{N}(\mu_X, \sigma_X^2)$	The Gaussian distribution with mean μ_X and variance σ_X^2

The Wiener-Khinchin theorem establishes a relationship between the autocorrelation function (ACF) and the power spectral density (PSD) for a wide-sense stationary (WSS) process. The formulas are given by:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} \mathrm{d}\tau, \quad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} \mathrm{d}f.$$

Additionally, when an output process $Y(t)$ passes through a linear time-invariant (LTI) system with frequency response $H(f)$, the PSD of the output is related to the input PSD by:

$$S_Y(f) = |H(f)|^2 S_X(f).$$

2.2 Phase noise and clock jitter

The phase shift of the clock signal brought about by the effect of noise in the circuit is known as excess phase $\Phi_e(t)$, which is a continuous-time stochastic process. If the double-sideband PSD of $\Phi_e(t)$ is $S_{\Phi_e}(f)$, the phase noise $\mathcal{L}(f)$ is defined as:

$$\mathcal{L}(f) := S_{\Phi_e}(f) \quad f \geq 0, \tag{1}$$

where f represents the offset from the carrier frequency, rather than the absolute frequency.

Clock jitter is defined as the time difference between the actual occurrence of an edge transition and its ideal occurrence time, which can be described as a discrete-time stochastic process. Assuming that T_0 denotes the average period of the clock signal and t_k where $k = 1, 2 \cdots n$ represents the moments when the clock's rising edge occurs, $\{J_a(k) := t_k - T_0\}$ represents the absolute jitter with its variance $\sigma^2_{J_a}$, $\{J_{p(N)}(k) := t_{k+N} - t_k - NT_0\}$ represents the N-period jitter with its variance $\sigma^2_{J_{p(N)}}$. Eq. (2) and (3) summarizes the relationship between variance of jitter and phase noise:

$$\sigma^2_{J_a} = \frac{2}{\omega_0^2} \int_0^{+\infty} \mathcal{L}(f)\mathrm{d}f, \tag{2}$$

$$\sigma^2_{J_{p(N)}} = \frac{8}{\omega_0^2} \int_0^{+\infty} \mathcal{L}(f)\sin^2(\pi f N/f_0)\mathrm{d}f, \tag{3}$$

where $\omega_0 = 2\pi/T_0$ and $f_0 = 1/T_0$ represent the average angular frequency and average frequency respectively. Specifically, if the accumulated cycle count N tends to infinity, and $\sigma^2_{J_{p(N)}}$ converges to a finite value, this value is defined as the long-term jitter variance, denoted as $\sigma^2_{J_{lt}}$: $\lim_{N\to\infty} \sigma^2_{J_{p(N)}} = \sigma^2_{J_{lt}}$.

2.3 Ornstein-Uhlenbeck process

The Ornstein-Uhlenbeck process is a special Gaussian process with mean-reversion properties and Markovianity. The Ornstein-Uhlenbeck process is described by the following stochastic differential equation (SDE):

$$dX(t) = \theta(\mu - X(t))dt + \sigma \mathrm{d}W(t), \tag{4}$$

where $X(t)$ follows Ornstein-Uhlenbeck process, μ is the long-term mean, $\theta > 0$ is the mean-reversion rate, σ is the volatility intensity, $W(t)$ is a standard Wiener process. At any finite time t, the distribution of $X(t)$, given $X(0) = x_0$, is Gaussian distribution:

$$X(t) \sim \mathcal{N}\left(\mu + (x_0 - \mu)e^{-\theta t}, \frac{\sigma^2}{2\theta}\left(1 - e^{-2\theta t}\right)\right). \tag{5}$$

As $t \to \infty$, the Ornstein-Uhlenbeck process reaches a stationary distribution, which is a Gaussian distribution with mean μ and variance $\frac{\sigma^2}{2\theta}$: $X(\infty) \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{2\theta}\right)$. At this time, the ACF $R_X(\tau)$ is exponentially decaying with the time lag τ:

$$R_X(\tau) = \frac{\sigma^2}{2\theta}e^{-\theta\tau}. \tag{6}$$

2.4 Principle of PLL-TRNG

Fig. 2 shows the basic PLL-TRNG architecture. Ideally, the D flip-flop (DFF) produces deterministic outputs with period $T_P = K_D T_0 = K_M T_1$, where K_D and K_M are constants and T_0 and T_1 represent the periods of clk_0 and clk_1. Realistic implementations exhibit clock jitter in both clk_0 and clk_1, creating uncertainty in the high-level state duration per T_P period. The XOR and 1-bit counter, work as a digitization step and its output is regarded as the raw random number (RRN), denoted as R_P. Notably, the XOR operation and 1-bit counter process analogue noise signals, rather than functioning as conditioning components, which are widely used in PLL-TRNGs and other types of TRNGs [SMS07, LBGD13, PV22]. This equivalence arises from XORing jitter signals with deterministic signals, without increasing output entropy. The high resolution of coherent sampling enables the generated RRN to achieve most full entropy. Consequently, PLL-TRNGs generally require no conditioning components for post-processing. The output bit rate R of the random number generator is given by $R = 1/T_P$. The sensitivity to jitter S is given by $S = \Delta^{-1}$ where $\Delta = T_1/K_D$ represents the time resolution within the reconstruction period T_1.

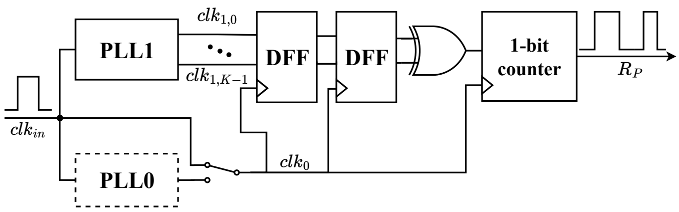


Figure 2: Basic architecture of PLL-TRNG

Fig. 3 illustrates the typical architecture of PLL in FPGA, which consists of a phase frequency detector(PFD), a charge pump(CP), a loop filter(LF), a voltage-controlled oscillator(VCO), and programmable configurable frequency dividers. The PFD and CP together form the phase detector(PD). M, N, and C_0 to C_{K-1} represent the multiplication factor, the input division factor, and the output division factors, respectively. For a PLL-TRNG using two PLLs, the designer needs to give a set of PLL configuration parameters $\zeta = \{M_0, N_0, C_0, M_1, N_1, C_1\}$, where $\{M_i, N_i, C_i\}, i = \{0, 1\}$ are the ones defined in Fig. 3.

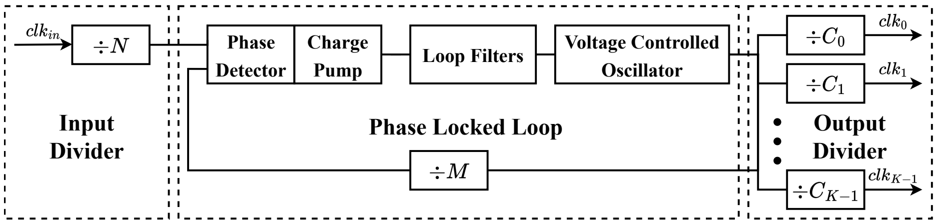


Figure 3: Basic architecture of PLL in FPGA

3 Phase noise model

3.1 Phase domain model of PLL

Phase noise in PLLs represents a widely researched topic, with linear approximation employed to derive the phase domain models [Cic03, XKGN09, DDS18]. When the sampling rate of the PFD is 10 times greater than the loop bandwidth, the PLL can be considered an LTI system [Cic03]. From this, we can derive the phase domain model of PLLs, as shown in Fig. 4, where K_{PD}, K_{VCO}, and $H_{LF}(s)$ represent the gains of the PD, the VCO, and the transfer function of the LF, respectively.

Each part of the PLL contributes to the phase noise $\Phi_{e,\cdot}(t)$. The stable operation of the PLL relies on negative feedback, which results in the output phase noise of the PLL being

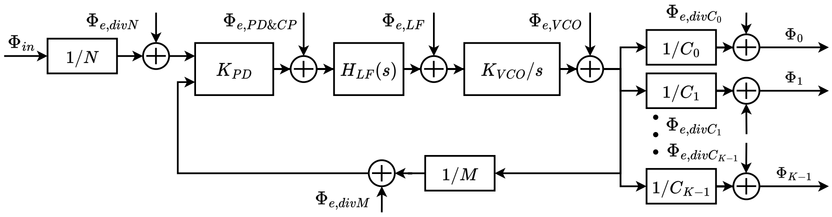


Figure 4: Phase domain model of PLL

more than a simple superposition of each component. Phase noise in the PLL is a complex issue that varies with the design parameters of the PLL. For TRNG designers, it is often impractical to fully grasp the design details of the PLL, as PLLs are typically provided as primitives within FPGAs. Therefore, it is essential to establish a simple, general, and easily applicable phase model.

We now analyze the PLL's phase noise components to develop a simplified phase noise model. The derivation begins with the PLL's closed loop transfer function:

$$H(s) := \frac{\Phi_{\text{VCO}}}{\Phi_{\text{in}}} = \frac{K_{PD}K_{VCO}H_{LF}(s)}{K_{PD}K_{VCO}H_{LF}(s) + s}. \tag{7}$$

Since the LF is a low-pass filter, $H(s)$ displays the low-pass feature. The PSD of PLL phase noise $\mathcal{L}_{\text{PLL}}$ is composed of two parts: the PSD of loop phase noise $\mathcal{L}_{\text{loop}}$ and the PSD of VCO phase noise $\mathcal{L}_{\text{VCO}}$ [XKGN09]. $\mathcal{L}_{\text{loop}}$ is the combination of input reference, PFD, CP, and feedback divider where white PM noise and flicker PM noise dominate. On the other hand, the white FM noise and flicker FM noise are the main contributions of $\mathcal{L}_{\text{VCO}}$ due to the integration effect of VCO. Therefore, $\mathcal{L}_{\text{loop}}$ and $\mathcal{L}_{\text{VCO}}$ can be given by $\mathcal{L}_{\text{loop}} = h_0 + h_1/f$ and $\mathcal{L}_{\text{VCO}} = h_2/f^2 + h_3/f^3$ [Cic03]. If the noise in the loop and the VCO are independent, $\mathcal{L}_{\text{PLL}}$ can be computed as:

$$\mathcal{L}_{\text{PLL}} = M^2|H(s)|^2\mathcal{L}_{\text{loop}} + |1 - H(s)|^2\mathcal{L}_{\text{VCO}}. \tag{8}$$

Recall that $H(s)$ is low-pass filtered, so $1 - H(s)$ is high-pass filtered. The contribution of loop and VCO phase noise to PLL phase noise varies with frequency. At low frequencies $\mathcal{L}_{\text{loop}}$ dominates while $\mathcal{L}_{\text{VCO}}$ is filtered by $|1 - H(s)|^2$, so $\mathcal{L}_{\text{PLL}}$ can be characterized as $\mathcal{L}_{\text{loop}}$. On the other hand, at high frequencies $\mathcal{L}_{\text{loop}}$ decays rapidly by $|H(s)|^2$, and $\mathcal{L}_{\text{PLL}}$ is primarily determined by $\mathcal{L}_{\text{VCO}}$. If we ignore the effect of flicker noise, $\mathcal{L}_{\text{PLL}}$ can be characterized as a flat band at low frequency and $1/f^2$ shape at high frequency, which can be approximated using the Lorentz spectrum [DDS18]:

$$\mathcal{L}_{\text{PLL}} \approx \frac{\mathcal{L}_0}{1 + (f/f_{\text{-3dB}})^2}, \tag{9}$$

where $\mathcal{L}_0 = \lim_{s\to 0} M^2|H(s)|^2\mathcal{L}_{\text{loop}} = M^2h_0$ is the flat band strength and $f_{\text{-3dB}}$ is the -3dB bandwidth of PLL. Fig. 5(a) shows the PLL phase noise based on the Lorentz spectrum assumption.

The main advantage of using this approximation is that there is no need to know the internal design parameters of the PLL (e.g. damping coefficients or natural frequency), and the phase noise of the PLL is only portrayed by the PLL's bandwidth, which is a readily adjustable parameter in FPGA implementations.

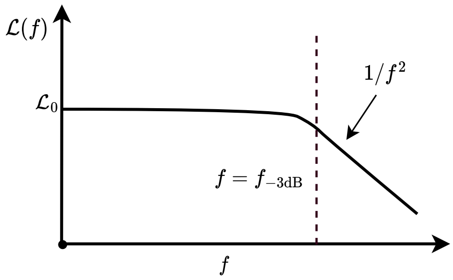
3.2 Jitter variance study

Based on the assumption that the output frequency divider is ideal, C_k-output frequency divider leads to the fact that the output absolute jitter of PLL is the same as the output absolute jitter of VCO, downsampled by the factor C_k [DDS18]. Therefore, when analyzing the PLL jitter, we can focus on the VCO output. According to Eq. (2), (3) and (9), we can derive the expression for the variances of absolute jitter J_a and N-period jitter $J_{p(N)}$:

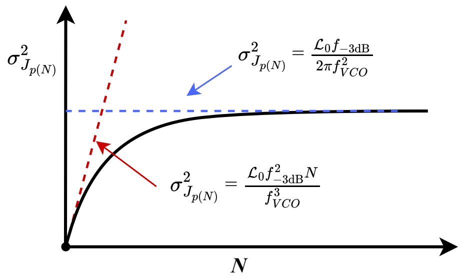
$$\sigma^2_{J_a} = \frac{\mathcal{L}_0 f_{\text{-3dB}}}{4\pi f^2_{\text{VCO}}}, \tag{10}$$

$$\sigma^2_{J_{p(N)}} = \frac{\mathcal{L}_0 f_{\text{-3dB}}}{2\pi f^2_{\text{VCO}}}\left(1 - \exp(-\frac{2\pi f_{\text{-3dB}} N}{f_{\text{VCO}}})\right), \tag{11}$$

where f_{VCO} is the frequency of VCO.



(a) Simplified phase noise spectrum



(b) N-period jitter variance

Figure 5: Phase noise and N-period jitter characteristics of the PLL

As shown in Fig. 5(b), $\sigma^2_{J_{p(N)}}$ exhibits two distinct asymptotic behaviors, corresponding to the cumulative and flat regions. The saturated value is the long-term jitter variance $\sigma^2_{J_{lt}}$, and is given by:

$$\sigma^2_{J_{lt}} = \frac{\mathcal{L}_0 f_{\text{-3dB}}}{2\pi f^2_{\text{VCO}}}. \tag{12}$$

3.3 Phase noise process

When the PLL is in the locked state, the output phase aligns with the input phase due to negative feedback. The output of the PLL is stationary when only thermal noise is considered [Meh02]. Therefore, the excess phase $\Phi_e(t)$ is a stationary process that satisfies the condition of the Wiener-Khinchin theorem. By applying the inverse Fourier transform, we can obtain the ACF of the phase noise $R_{\Phi_e}(\tau)$:

$$R_\Phi(\tau) = \mathcal{L}_0 \pi f_{\text{-3dB}} e^{-2\pi f_{\text{-3dB}}|\tau|}. \tag{13}$$

It is observed that the expressions in Eq. (13) and Eq. (6) are quite similar. If we use a first-order approximation to inscribe the phase noise of the PLL with the Lorentz spectrum, we can model the excess phase of the PLL as an Ornstein-Uhlenbeck process with $\mu = 0$, $\theta = 2\pi f_{\text{-3dB}}$ and $\sigma = 2\pi f_{\text{-3dB}}\sqrt{\mathcal{L}_0}$. It is defined by the following SDE:

$$\mathrm{d}\Phi_e(t) = 2\pi f_{\text{-3dB}}(-\Phi_e(t)dt + \sqrt{\mathcal{L}_0}\mathrm{d}W(t)). \tag{14}$$

Eq.(14) shows the characterization of the output phase of a PLL. When the PLL is not locked, its phase has a mean-reversion property, and the rate of regression depends on the loop bandwidth. When the PLL is locked, its phase has a stationary property, which indicates that the phase at any two moments after locking is identically distributed. We assume that when the PLL is locked, $\Phi_e(0)$ already has the same distribution as the stationary state, and the output excess phase has is WSS with Gaussian distribution:

$$\Phi_e(t) \sim \mathcal{N}\left(0, \mathcal{L}_0 \pi f_{\text{-3dB}}\right). \tag{15}$$

4 Enhanced stochastic model

Inspired by the work presented in [PV24b], the joint distribution of the excess phase across multiple PLL outputs within a pattern period T_P can be effectively modeled utilizing a multivariate Gaussian distribution. To simplify the model, we assume that clk_0 is jitter-free and only clk_1 exhibits clock jitter.

4.1 Excess phase process for multiple PLL outputs

The K outputs of PLL1, denoted as $clk_{1,0}, clk_{1,1}, \ldots, clk_{1,K-1}$, are utilized. Assuming all outputs exhibit a duty cycle of α, and the initial phase between $clk_{1,0}$ and clk_0 is ϕ_0. As all outputs derive from the same PLL, each output represents the same stochastic process but with distinct delays. The delays for these K outputs are defined as $0, \tau, \ldots, (K-1)\tau$. Within a pattern period T_P, all outputs of PLL1 are sampled K_D times by clk_0. For the i-th sample, the phase and excess phase of $clk_{1,k}$ $(0 \le k \le K-1)$ is expressed as $\Phi(i,k) := \Phi(t_i - k\tau)$ and $\Phi_e(i,k) := \Phi_e(t_i - k\tau)$, where $t_i = i \times T_0$, $0 \le i \le K_D - 1$ is the sampling instant.

To facilitate the modeling of the excess phase, we arrange the excess phases at all sampling instants within one T_P period into a $K_D \times K$ dimensional matrix $\mathbf{\Phi_e}$, which is given by:

$$\mathbf{\Phi_e} = \begin{bmatrix} \Phi_e(t_0), & \Phi_e(t_0-\tau), & \ldots, & \Phi_e(t_0-(K-1)\tau), \\ \Phi_e(t_1), & \Phi_e(t_1-\tau), & \ldots, & \Phi_e(t_1-(K-1)\tau), \\ \vdots & \vdots & \vdots & \vdots \\ \Phi_e(t_{K_D-1}), & \Phi_e(t_{K_D-1}-\tau), & \ldots, & \Phi_e(t_{K_D-1}-(K-1)\tau) \end{bmatrix}. \tag{16}$$

Based on the same method, we can also construct the phase matrix $\mathbf{\Phi}$.

Example 1 Let $K_M = 4$, $K_D = 5$, $K = 2$ and $\tau = T_1/4$. The process of modeling the excess phase matrix $\mathbf{\Phi_e}$ and the phase matrix $\mathbf{\Phi}$ is shown in Fig. 6.

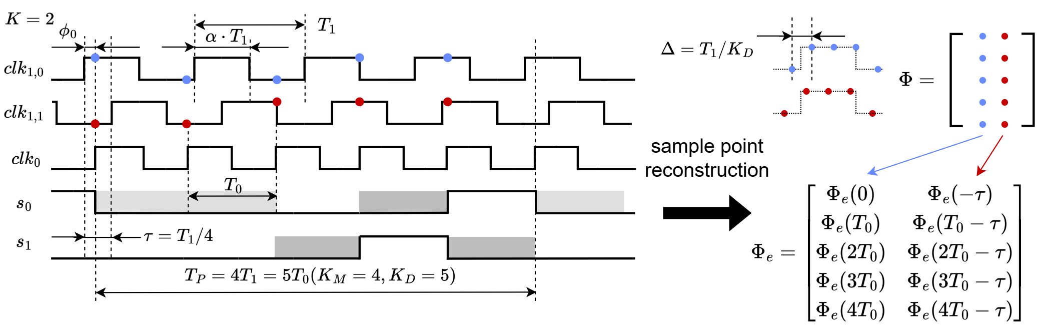


Figure 6: Principle of excess phase modeling, for the case of $K = 2$ outputs, with $K_D = 5$, $K_M = 4$ and $\tau = T_1/4$.

4.2 PLL-TRNG stochastic model based on Gaussian process

While existing work neglects sampling point correlations [FBB19, FBB+23], the enhanced stochastic model provides quantitative characterization and yields improved entropy lower bounds. We first define the phase set $\mathcal{P}$, representing all phases where clk_1 is high:

$$\mathcal{P} := \{\phi \mid kT_1 < \phi < (k+\alpha)T_1,\, k \in \mathbb{Z}\}. \tag{17}$$

We use $\chi_{\mathcal{P}}(x)$ to represent the indicator function of the set $\mathcal{P}$, which indicates whether the element x belongs to the set $\mathcal{P}$:

$$\chi_{\mathcal{P}}(x) := \begin{cases} 1, & \text{if } x \in \mathcal{P} \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

Clock jitter is a discrete-time random process that can be regarded as the sampling of the excess phase. $-\Phi_e(i,k)/\omega_0$ converts the excess phase to the absolute jitter of the i-th sample for the k-th output. According to [FBB+23], the random variable $\Phi(i,k) = \phi_0 - k\tau + j(i) \times \Delta - \Phi_e(i,k)/\omega_0$ represents the phase of the i-th sampled bit. Here, $\Delta = T_1/K_D$ represents the time resolution of coherent sampling, and $j(i) = i \times K_M \bmod K_D$ represents the index of the sampled bit in the reconstructed period. The conditional PMF, given the excess phase, equals to:

$$f_{B_{i,k}|\Phi_e(i,k)}(b \mid \varphi) = \begin{cases} \chi_{\mathcal{P}}\left(\Phi(i,k)\right), & \text{if } b = 1 \\ 1 - \chi_{\mathcal{P}}\left(\Phi(i,k)\right), & \text{if } b = 0 \end{cases} \tag{19}$$

Eq. (19) shows that any two sampling points are conditionally independent. Considered sampling points $B_{i,k}$ with index $\forall i \in \{0, 1, \cdots, K_D - 1\}$ and $\forall k \in \{0, 1, \cdots, K-1\}$, we define the bit matrix $\boldsymbol{B} := [B_{i,k}]_{K_D \times K}$, with PMF equal to:

$$f_{\boldsymbol{B}|\boldsymbol{\Phi}_e}(\boldsymbol{b} \mid \boldsymbol{\varphi}) = \prod_{i=0}^{K_D-1} \prod_{k=0}^{K-1} f_{B_{i,k}|\Phi_e(i,k)}(b_{i,k} \mid \varphi_{i,k}). \tag{20}$$

Once we find the conditional PMF of the bit matrix $\boldsymbol{B}$, the joint PMF of $\boldsymbol{B}$ equal to:

$$f_{\boldsymbol{B}}(\boldsymbol{b}) = \int_{\mathbb{R}^{K_D \times K}} f_{\boldsymbol{B}|\boldsymbol{\Phi_e}}(\boldsymbol{b} \mid \boldsymbol{\varphi}) f_{\boldsymbol{\Phi_e}}(\boldsymbol{\varphi}) \mathrm{d}\boldsymbol{\varphi}, \tag{21}$$

with the excess phase matrix $\boldsymbol{\Phi_e}$ follows $K_D \times K$-dimensional multivariate normal distribution. The conditioned PMF for the raw random number R_p, given the sampled bit matrix $\boldsymbol{B}$, is equal to:

$$f_{R_p|\boldsymbol{B}}(r \mid \boldsymbol{b}) = \begin{cases} 1, & \text{if } r = \bigoplus_{0 \le i \le K_D - 1} \bigoplus_{0 \le k \le K-1} b_{i,k} \\ 0, & \text{otherwise} \end{cases} \tag{22}$$

The unconditional PDF of R_p can be computed by Eq. (22), which equals to:

$$f_{R_p}(r) = \sum_{\boldsymbol{b} \in \{0,1\}^{K_D \times K}} f_{R_p|\boldsymbol{B}}(r \mid \boldsymbol{b}) \cdot f_{\boldsymbol{B}}(\boldsymbol{b}). \tag{23}$$

According to Eq. (19) to (23), we can derive the expression for the distribution of R_p based on the $\boldsymbol{\Phi_e}$:

$$
\begin{aligned}
f_{R_p}(r) &= \sum_{\boldsymbol{b}\in\{0,1\}^{K_D\times K}} f_{R_p|\boldsymbol{B}}(r \mid \boldsymbol{b}) \int_{\mathbb{R}^{K_D\times K}} f_{\boldsymbol{B}|\boldsymbol{\Phi_e}}(\boldsymbol{b}\mid\boldsymbol{\varphi}) f_{\boldsymbol{\Phi_e}}(\boldsymbol{\varphi})\mathrm{d}\boldsymbol{\varphi} \\
&= \int_{\mathbb{R}^{K_D\times K}} \left[\sum_{\substack{\boldsymbol{b}\in\{0,1\}^{K_D\times K}\\ r=\oplus_{i,k} b_{i,k}}} f_{\boldsymbol{B}|\boldsymbol{\Phi_e}}(\boldsymbol{b}\mid\boldsymbol{\varphi}) \right] f_{\boldsymbol{\Phi_e}}(\boldsymbol{\varphi})\mathrm{d}\boldsymbol{\varphi} \\
&= \frac{1}{2}\left(1 + (-1)^r \int_{\mathbb{R}^{K_D\times K}} \prod_{i=0}^{K_D-1}\prod_{k=0}^{K-1} \left(1 - 2\chi_{\mathcal{P}}\left(\Phi(i,k)\right)\right) f_{\boldsymbol{\Phi_e}}(\boldsymbol{\varphi})\mathrm{d}\boldsymbol{\varphi}\right).
\end{aligned} \tag{24}
$$

Eq. (24) generalizes Proposition 3 in [FBB+23] and remains valid regardless of the technique to control the distance in time between contributors described in [FBB+23].

The Shannon entropy H_1 and min-entropy H_∞ of R_p can be computed as follows:

$$H_1 := -f_{R_p}(1)\log_2 f_{R_p}(1) - f_{R_p}(0)\log_2 f_{R_p}(0), \tag{25}$$

$$H_\infty := -\log_2(\max(f_{R_p}(0), f_{R_p}(1))). \tag{26}$$

4.3 Correlation study

Correlation between sampling points limits the entropy of PLL-TRNGs, which is especially severe in the multi-output structure. The enhanced stochastic model quantifies the correlation and identifies its key influencing factors.

4.3.1 Correlation between excess phase

According to Eq.(13), the correlation of excess phase $\rho_\Phi(t_m, t_n)$, given two specific time t_m and t_n is given by:

$$
\begin{aligned}
\rho_\Phi(t_m, t_n) :&= \frac{\mathbf{Cov}(\Phi_{t_m}, \Phi_{t_n})}{\sqrt{\mathbf{Var}(\Phi_{t_m})\mathbf{Var}(\Phi_{t_n})}} \\
&= \exp(-2\pi f_{\text{-3dB}}\left|t_m - t_n\right|),
\end{aligned} \tag{27}
$$

where we find that $\rho_\Phi(t_m, t_n)$ is a function of $f_{\text{-3dB}}$. The larger the bandwidth, the faster the excess phase autocorrelation coefficient decays with time.

4.3.2 Correlation between sampling points

In the design of PLL-TRNG, the entropy is mainly affected by the contributors [FBB+23]. Therefore, it is important to analyze the correlation of the contributors. The correlation between sampling points B_{m,k_m} and B_{n,k_n} equals to:

$$\rho_B(B_{m,k_m}, B_{n,k_n}) := \frac{\mathbf{Cov}(B_{m,k_m}, B_{n,k_n})}{\sqrt{\mathbf{Var}(B_{m,k_m})\mathbf{Var}(B_{n,k_n})}}, \tag{28}$$

where the covariance and variance can be computed by Eq. (21).

4.4 Model simulation

According to Eq. (27) and (28), we can use Monte Carlo integration to simulate the correlation of excess phase and contributors within the pattern period T_P. All results in this work, employed the `numpy.random.multivariate_normal` (version 1.26.4) method to generate one million random samples for each experiment.

4.4.1 Correlation of excess phase and contributors

Since the correlation of excess phase is the function of the bandwidth of PLL and time lag, we can deduce that when the PLL configuration is given, the correlation of contributors is also the function of both.

The effect of the noise intensity $\mathcal{L}_0$ versus the bandwidth $f_{\text{-3dB}}$ on bit correlation is shown in the Fig. 7 and 8 respectively. It can be observed that for a given PLL configuration, the bit correlation is related to the bandwidth of the PLL and not to the noise intensity $\mathcal{L}_0$. Consistent with phase correlation, increasing the bandwidth $f_{\text{-3dB}}$ causes bit correlation to decay more quickly. Although increasing the noise intensity $\mathcal{L}_0$ leads to an increase in the number of contributors, it does not change the correlation of sample bits for a given time lag.

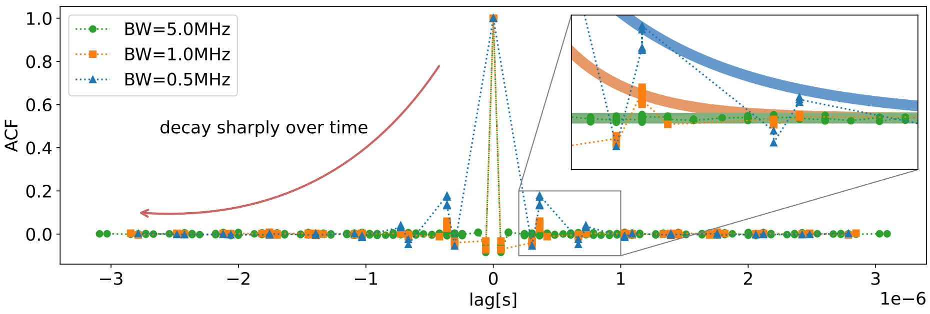


Figure 7: Relationship between bit correlation and time with PLL configuration $\zeta = \{53, 5, 7, 28, 3, 3\}$ for bandwidths of 0.5MHz, 1MHz and 5MHz.

In addition, it can be found in Fig. 8 that the bit correlation is generally smaller than the phase correlation:

$$\rho_B(B_{m,k_m}, B_{n,k_n}) \leq \rho_{\Phi_e}(t_m, t_n),$$

and thus we can estimate the bit correlation based on passing through the phase correlation. In the design process, our goal is to maximize the distance between contributors, which is measured in terms of T_0 cycles. If the correlation of sample bits is below a certain threshold $\rho_{B,\max}$, the corresponding minimal distance between sample bits $d_{\min}$ can be calculated as:

$$d_{\min} \geq \frac{-\ln \rho_{B,\max}}{2\pi} \cdot \frac{f_0}{f_{\text{-3dB}}}. \tag{29}$$

The designer can estimate this threshold ahead of the hardware implementation and optimize the PLL configuration based on this threshold.

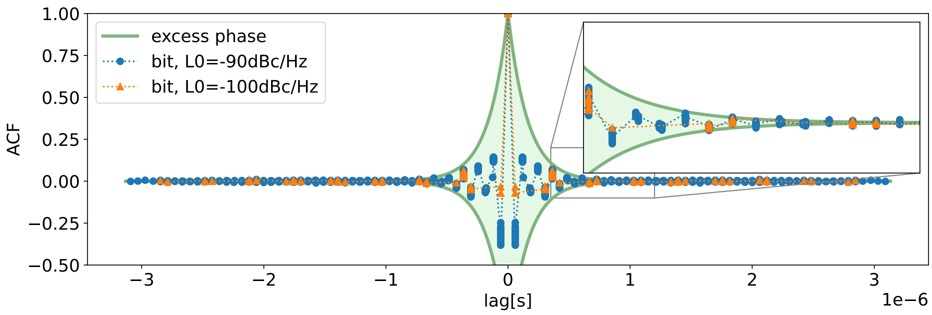


Figure 8: Relationship between bit correlation and time with PLL configuration $\zeta = \{53, 5, 7, 28, 3, 3\}$ for $\mathcal{L}_0$ of -90dBc/Hz and -100dBc/Hz.

Example 2 Given the following PLL configuration parameters $f_{\text{-3dB}} = 4\text{MHz}$, $f_0 = 400\text{MHz}$ and the maximum correlation of sample bits $\rho_{B,\max} = 0.001$, it can be obtained $d_{\min} = 109$. If $\rho_{B,\max} = 0.01$ and $\rho_{B,\max} = 0.1$, then we have $d_{\min} = 73$ and $d_{\min} = 37$.

4.4.2 Effect of Bandwidth on entropy

In Sect. 4.4.1, we can conclude that the bandwidth of PLL affects the correlation of both excess phase and contributors. We can further discuss its effect on entropy. According to Eq. (10) and (28), an increase in bandwidth leads to a larger jitter variance, while making the correlation decay more rapidly over time. Both effects are favorable for entropy improvement. Fig. 9 shows the effect of the PLL bandwidth $f_{\text{-3dB}}$ on entropy. We demonstrate that optimal bandwidth selection can improve the entropy and this finding necessitates the explicit inclusion of PLL bandwidth as a critical design parameter.

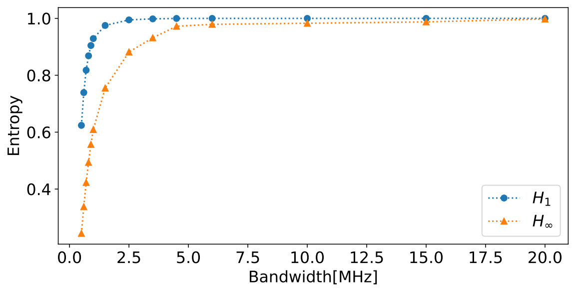
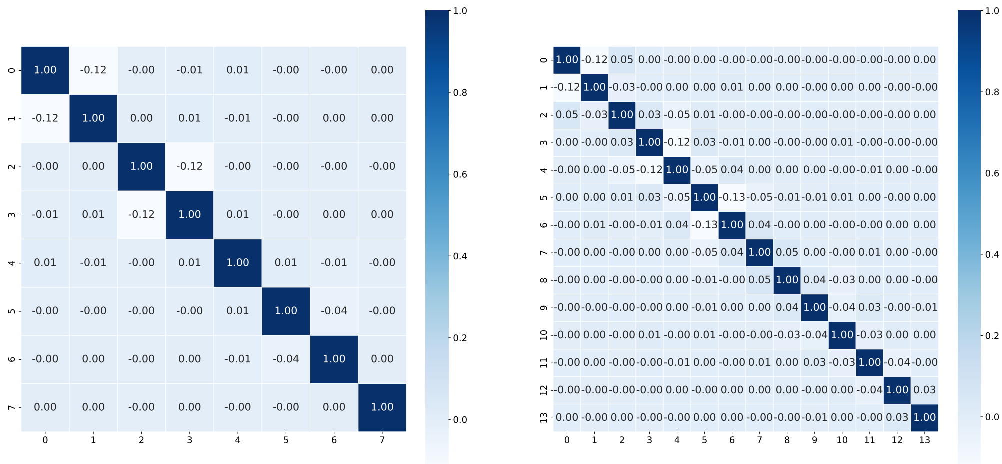


Figure 9: Min-entropy and Shannon entropy of a single output PLL-TRNG for a given PLL configuration $\zeta = \{49, 4, 8, 16, 1, 3\}$ with a change in bandwidth from 0.5 MHz to 20 MHz.

4.4.3 Effect of PLL outputs on correlation and entropy

Our enhanced stochastic model characterizes both the bit correlations and entropy in multi-output configurations.

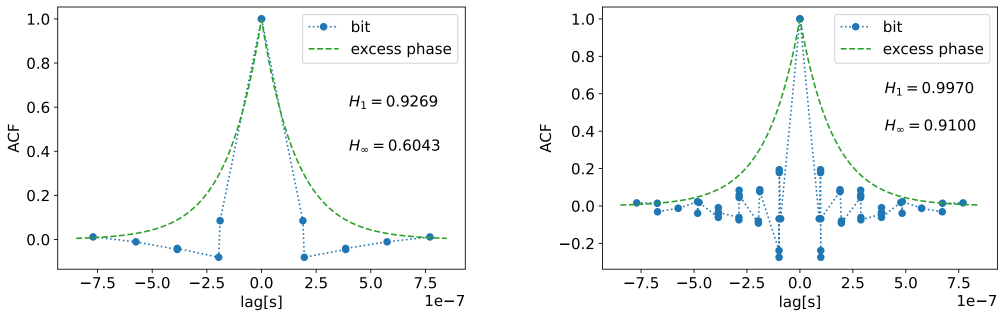


(a) Single PLL output (b) Two PLL outputs

Figure 10: Given a PLL configuration $\zeta = \{49, 4, 8, 16, 1, 3\}$, the correlation matrix of contributors for single PLL output (left) and two PLL outputs (right).

Fig. 10 and Fig. 11 show the covariance matrix of contributors and the relationship between bit correlation and time lag for single and dual PLL1 outputs, respectively. The min-entropy H_∞ and Shannon entropy H_1 lower bound for both cases are shown in Fig. 11.

While additional PLL outputs provide more contributors, they simultaneously introduce higher correlations. If the benefit of additional contributors outweighs the increased bit correlation, then carefully increasing PLL outputs can enhance entropy generation. Notably, bit correlation remains consistently lower than phase correlation across all PLL output configurations.



(a) Single PLL output (b) Two PLL outputs

Figure 11: Given a PLL configuration $\zeta = \{49, 4, 8, 16, 1, 3\}$, the relationship between bit correlation and time lag for single PLL output (left) and two PLL outputs (right).

4.5 Discussion on assumptions for the stochastic model

The stochastic model in [FBB+23] relies on four key assumptions. Our proposed stochastic model establishes the theoretical foundation for each of these assumptions.

Assumption 1 (Stationarity of the pattern) and Assumption 2 (Output stationarity), can be explained in terms of asymptotic stationarity of the Ornstein-Uhlenbeck process. Because the phase noise of the PLL can be regarded as a wide stationary process when the PLL is locked, the output bit is also stationary. Stationarity is a significant advantage of PLL due to other jitter-based entropy sources.

Assumption 3 (Small intra-pattern correlations) can be theoretically justified through the ACF of the Ornstein-Uhlenbeck process. The ACF of the excess phase decays exponentially with a time lag. Section 4.4 demonstrates that bit correlation remains consistently weaker than phase correlation. This permits the application of an exponential decay model to estimate the bit correlation upper bound, thereby ensuring weakly correlated contributors within the pattern period T_P. We can also quantitatively estimate the lower bound on the time lag of the sample points for a given correlation threshold. This method enables a priori control of sampling point correlations before hardware realization, thereby serving as an optimization constraint for PLL configuration algorithms.

Assumption 4 (Independence of least significant bits of the counter output) derives from Assumption 3. When the sampling point correlations are sufficiently weak, the raw random bits can be treated as statistically independent. However, we emphasize that the number of PLL outputs must be carefully constrained, as excessive outputs amplify both intra pattern bit correlation and raw random number correlation.

5 Design and implementation

5.1 Design of the PLL-TRNG

Overall design Fig. 12 presents the overall design block diagram of the PLL-TRNG. The digital noise source is used to generate raw random numbers. We use a design with two PLLs to make the PLL configuration more flexible. The jitter measurement module measures the PLL clock jitter through a time-to-digital converter composed of a cascaded CARRY4 primitives and an asynchronous counter. We estimate the entropy of the raw random numbers for AIS-31 [KS11] and NIST [TBK+18] standards.

The structure of the designed PLL-TRNG circuit is shown in Fig. 2. We choose a circuit structure with two PLLs, where PLL1 has two outputs with a phase difference of $\pi/2$. The input frequency of both PLLs is set to 100MHz, while the PLL bandwidth is configured to 4MHz to increase the entropy rate of the PLL-TRNG.

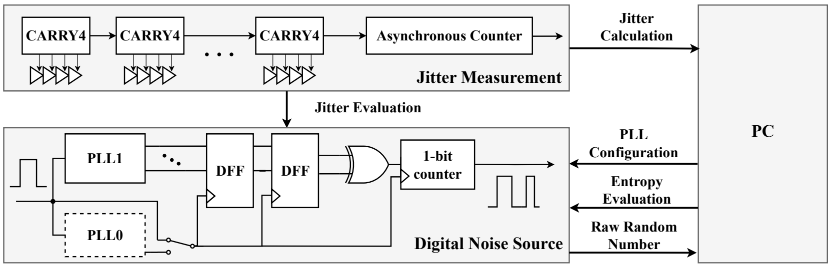


Figure 12: Overall design of PLL-TRNG

PLL setup We use the Python script in [FBB+23] to optimize the PLL configuration ζ. We select the bit correlation threshold $\rho_{\max} = 0.1\%$ when performing the configuration search, and according to Sect. 4.4.1, we can compute the minimum distance between contributors $d_{\min}$. Combining the jitter sensitivity S, throughput R and the average distance between contributors d_{mean}, the optimal configuration parameter ζ is obtained as shown in Table 2. In this configuration, the parameter $S = 0.1376\text{ps}^{-1}$, $R = 0.397\text{kb/s}$, while all the distance between contributors is greater than $d_{\min}$, which guarantees the independence of the raw random numbers.

Table 2: Optimal PLL configuration parameter ζ

M_0	N_0	C_0	M_1	N_1	C_1	K_M	K_D	R[MHz]	S[ps^{-1}]	$d_{\min}$	d_{mean}
43	4	7	32	3	3	896	387	0.397	0.1376	56.28	55.29

5.2 Differential on-chip jitter measurement

5.2.1 Basic principle

Method We adopt the differential on-chip jitter measurement technique as outlined in [YRG+17] to assess the jitter of the PLL output. This method effectively mitigates global jitter effects through differential measurement, offering high accuracy and ease of FPGA implementation [BGH+24]. It has been widely implemented in FPGA [YRG+17] and ASIC [PV24a] designs.

At its core, the technique involves the use of a time-to-digital converter (TDC) to determine the phase of the PLL output clock. The TDC consists of a delay chain (DC) and an asynchronous counter (ACNT), where the DC provides precise time measurements and the ACNT tracks the number of complete clock cycles. By combining both measurements, the PLL clock phase is accurately determined.

Circuit structure The circuit for jitter measurement is depicted in Fig. 13. Since the jitter measurement is performed differentially, PLL0 and PLL1 are configured identically. Upon PLL reset, a lock-up period is required for stabilization, after which the PLL output becomes stable. The `locked` signal controls the enable input of the DC, ensuring that only data collected after PLL lock is captured. Each DC consists of 256 stages, implemented using cascaded CARRY4 primitives, for a total of 64 primitives. The output of each stage is connected to a D flip-flop, which captures the DC data under the control of the shared `dc_clk` signal. The ACNT is implemented using a cascade of 32 T flip-flops. Data from both DCs and ACNTs (a total of 576 bits) are transmitted serially to the PC for subsequent processing. The measurement process is repeated $N_{\text{exp}} = 5000$ times, with the phase difference computed for each iteration.

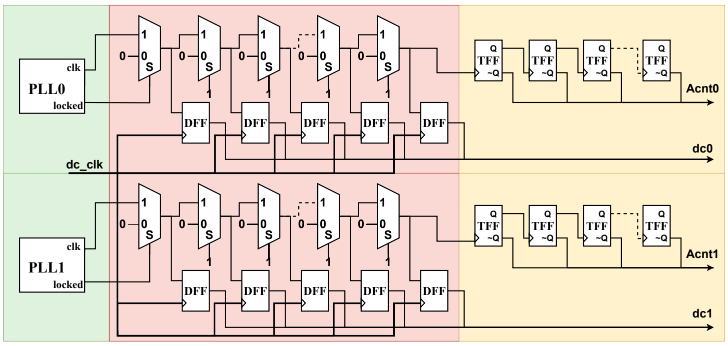


Figure 13: Circuit architecture of jitter measurement

5.2.2 Phase variance model

To start, we consider how the phase of clk changes over time with jitter. Suppose the phase of the PLL clock clk_i ($i = \{0, 1\}$) at time t_0 is Φ^i_{start}. Then, at time $t_0 + t$, the phase Φ^i_{stop} of clk_i can be defined as:

$$\Phi^i_{\text{stop}}(t) = 2\pi f^i_n t + \Phi^i_e(t) + \Phi_g(t) + \Phi^i_{\text{start}}, \tag{30}$$

where f^i_n represents the average frequency of clk_i, $\Phi^i_e(t)$ represents the excess phase due to local jitter of clk_i, and $\Phi_g(t)$ represents the excess phase caused by global noise (e.g., power supply variations). Thus the phase difference $\Phi^i_\Delta(t)$ of clk_i is given by:

$$\Phi^i_\Delta(t) = \Phi^i_{\text{stop}}(t) - \Phi^i_{\text{start}} = 2\pi f^i_n t + \Phi^i_e(t) + \Phi_g(t), \tag{31}$$

which yields:

$$\frac{\Phi^0_e(t)}{2\pi f^0_n} - \frac{\Phi^1_e(t)}{2\pi f^1_n} = \frac{\Phi^0_\Delta(t)}{2\pi f^0_n} - \frac{\Phi^1_\Delta(t)}{2\pi f^1_n} + \frac{\Phi_g(t)}{2\pi}\frac{(f^1_n - f^0_n)}{f^0_n f^1_n}. \tag{32}$$

To neglect the effect of $\Phi_g(t)$, the differential jitter measurement requires that the frequency difference between clk_0 and clk_1 be very small. This is easily achievable for PLLs because the output frequency of a PLL can be configured through division factors. We assume $f^0_n = f^1_n = f_n$. At this point, the third term in Eq. (32) represents the relative N-period jitter $J^{f_n}_{r,p(N)}$ between clk_0 and clk_1 at frequency f_n [DDS18]. Defining $\Phi_\Delta = \Phi^0_\Delta - \Phi^1_\Delta$, we obtain:

$$J^{f_n}_{r,p(N)} = \frac{1}{2\pi f_n}(\Phi^0_\Delta - \Phi^1_\Delta) = \frac{\Phi_\Delta}{2\pi f_n}, \tag{33}$$

$$\sigma^2_{J^{f_n}_{r,p(N)}} = \frac{1}{(2\pi f_n)^2}\mathbf{Var}(\Phi_\Delta). \tag{34}$$

From Eq. (34), we can see that differential jitter measurement obtains the relative N-period jitter between two clock signals at the same frequency.

The differential jitter measurement results cannot be directly applied to the stochastic model of PLL-TRNG for two key reasons. First, the two PLLs in PLL-TRNG typically operate at different frequencies. As Eq. (12) indicates, their jitter characteristics differ, necessitating independent measurement and synthesis. Second, PLL-TRNG fundamentally relies on the relative absolute jitter between the two PLLs. Therefore, the relationship between relative N-period jitter and relative absolute jitter must be explicitly considered.

We consider both PLLs in the differential jitter measurement to have the same jitter and thus have $\sigma^2_{J^{f_n}_{p(N)}} = \frac{1}{2}\sigma^2_{J^{f_n}_{r,p(N)}}$ where $\sigma^2_{J^{f_n}_{p(N)}}$ represents the N-period jitter variance generated by the PLL operating at frequency f_n over N cycles. Assuming that during the operation of the PLL-TRNG, PLL0 and PLL1 operate at frequencies f_{n0} and f_{n1} respectively, to convert the clock jitter of both to the output clock of PLL0, we can use the method mentioned in [BLMT11]. The converted N-period jitter variance $\sigma^2_{p(N)}$ is given by:

$$\sigma^2_{J_{p(N)}} = \sigma^2_{J^{f_{n1}}_{p(N)}} + \frac{f_{n0}}{f_{n1}}\sigma^2_{J^{f_{n0}}_{p(N)}}. \tag{35}$$

The N-period jitter variance $\sigma^2_{J_{p(N)}}$ stabilizes with the long-term jitter variance $\sigma^2_{J_{lt}}$ if we ensure that the measurement time is long enough for the PLL jitter to saturate. At this point according to Eq. (10) and (12) we have:

$$\sigma^2_{J_a} = \frac{1}{2}\sigma^2_{J_{lt}}. \tag{36}$$

By measuring the differential jitter, we obtain the values of $\sigma^2_{J^{f_{n0}}_{p(N)}}$ and $\sigma^2_{J^{f_{n1}}_{p(N)}}$. Applying the jitter synthesis technique of Eq. (35) and (36), we can estimate the relative absolute jitter $\sigma^2_{J_a}$ between PLL0 and PLL1 in the hardware implementation.

5.2.3 Delay chain characterization

Characterization method The accuracy of the TDC is intrinsically dependent on the delays d_j within the DC. To ensure accurate differential jitter measurements, it is imperative to first characterize the individual delay units, as process variations and interconnect delays can introduce significant nonlinearity in the DC. The Monte Carlo method, as outlined in [YRG+17], is utilized to perform the DC characterization, as illustrated in Fig. 14. A ring oscillator generates an independent sampling clock `dc_clk` at a frequency of 5Hz, allowing for the accumulation of sufficient clock jitter over 10,000 samples, which are then processed via serial communication on a host PC.

Ring oscillator period measurement To support the characterization of the DC, the period T_{RO} of the Ring Oscillator (RO) is measured using the method detailed in [BGH+24]. Given a system clock period of $T_{\text{sys}} = 10$ns, the average RO period is determined to be $T_{RO} = 2.0358$ns. This measurement ensures the effectiveness of random sampling within the DC. The histogram of the obtained period values is presented in Fig. 15(a).

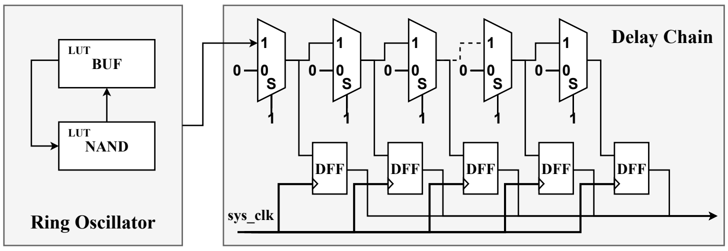
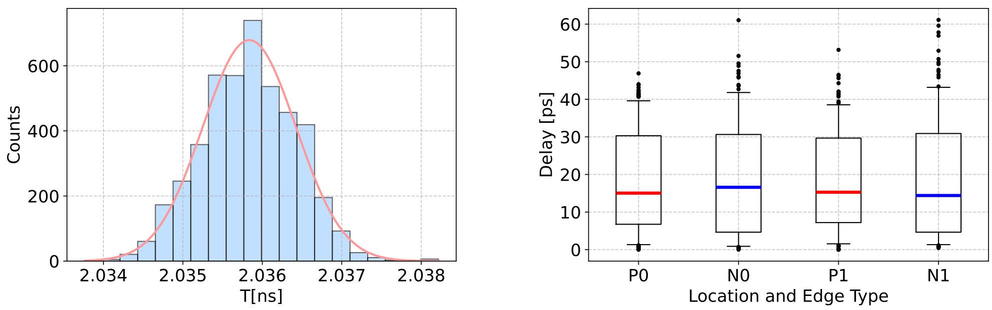


Figure 14: Circuit Architecture of Delay Chain Characterization

Characterization results Fig. 15(b) presents the results of the delay characterization for the rising-edge delay d_{01} and falling-edge delay d_{10} across 256-stage DCs at the X6Y60 and X10Y60 locations on a Xilinx Artix-7 FPGA. Both measurement locations exhibit comparable average delays, $\mu_d \approx 15$ps, and variances, $\sigma_d \approx 13$ps, which satisfy the precision requirements for jitter measurements, as specified in [MGBF+23].

Despite the satisfactory precision, the presence of zero-delay units within the DC introduces potential inaccuracies in the measurements due to clock skew and nonlinearity, which are characteristic of advanced technologies. To mitigate these issues, techniques such as bubble rearrangement have been employed to improve the linearity and overall measurement accuracy.



(a) RO period measurement with the mean period $T_{RO} = 2.0358$ ns, using 10000 samples.

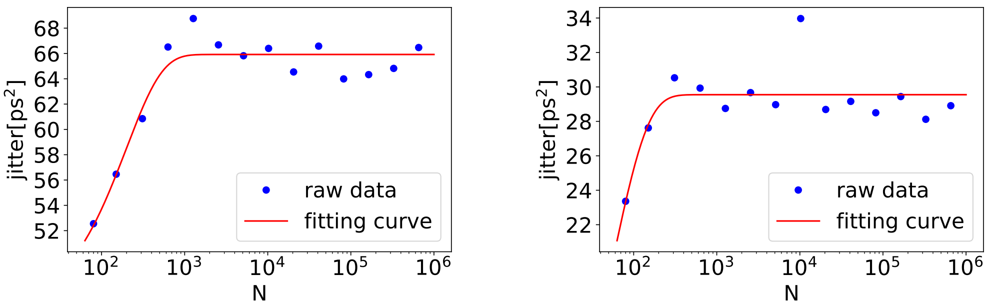
(b) DC delay distribution, using 10000 samples. The x-axis label show the data is for rising or falling edge (P- or N-) and for X6Y60 or X10Y60 (-0 or -1).

Figure 15: Delay Chain Characterization Results

5.3 Results of jitter measurement

N-period jitter measurement In this study, we employ a jitter measurement technique to assess the N-period jitter of both PLL0 and PLL1. The results, as depicted in Fig. 16, span a jitter accumulation time range from 100ns to 50μs. The experimental data show strong consistency with the theoretical expression in Eq. (12), thereby validating the phase noise model. From Fig. 16, we observe that some points deviate from the curve shown in Fig. 5(b). This is due to the simplification of Eq. (9), where we consider the PLL as a simple first-order system, however PLLs used in practice are usually second-order. This simplification does not affect the magnitude of the long-term jitter [Lee02]. Although the PLLs utilized in the FPGA are not strictly first-order systems, it is a reasonable approximation to model the PLL phase noise as a Lorentzian spectrum.

To quantify the magnitude of the jitter, the nonlinear least squares (NLS) fitting method is applied to the function $y = a \cdot (1 - \exp(-bx)) + c$. The experimental results indicate that the long-term jitter variances for PLL0 and PLL1 are as follows: $\sigma^2_{J_{lt}^{fn0}} = 20.21/2 = 10.105\,\mathrm{ps}^2$ and $\sigma^2_{J_{lt}^{fn1}} = 26.78/2 = 13.39\,\mathrm{ps}^2$. Notably, the quantization errors during the measurements of PLL0 and PLL1 differ significantly, with $Q_0 = 45.71\,\mathrm{ps}^2$ and $Q_1 = 2.77\,\mathrm{ps}^2$, respectively. This discrepancy may be attributed to the inherent nonlinearity of the DC.



(a) PLL0 with the configuration $\{M = 43, N = 4, C = 7\}$ with the fitting curve $y = 20.21(1 - \exp(-2.71x)) + 45.71$

(b) PLL1 with the configuration $\{M = 32, N = 3, C = 3\}$ with the fitting curve $y = 26.78(1 - \exp(-9.73x)) + 2.77$

Figure 16: N-period jitter $\sigma^2_{J_{p(N)}}$ of PLL0 and PLL1.

Equivalent relative jitter Both PLL0 and PLL1 exhibit clock jitter, and the PLL-TRNG system exploits the equivalent relative jitter between PLLs operating at distinct frequencies. Utilizing Eq. (35) and (36), the equivalent long-term jitter is computed to be $\sigma^2_{J_{lt}} = 23.574\,\mathrm{ps}^2$, which in turn yields the equivalent absolute jitter of $\sigma^2_{J_a} = 11.79\,\mathrm{ps}^2$. A detailed quantitative evaluation of the lower bound on the jitter is provided in Sect. 6.1.

6 Results and comparison

6.1 Entropy evaluation

Tests of NIST SP 800-90B and AIS-31 We conduct NIST SP 800-90B non-IID test and Test Suite AIS20/31 to verify the security of the PLL-TRNG. We collect 100Mb of raw random numbers, and the experimental results show that the random numbers pass all the tests of NIST SP 900-90B, obtaining a min-entropy of 0.9374. The data is also put through the Test Suite AIS20/31 test and passes all the T0-T7 tests while achieving full entropy in the T8 test.

Evaluation based on stochastic model We evaluate the entropy rate of the designed PLL-TRNG based on the stochastic model described in Sect. 4. The design parameters p_d and platform parameters p_p to be determined are first analyzed as shown in Table 3. For the platform parameters, the measurement method for the relative jitter variance $\sigma^2_{J_a}$ between PLL0 and PLL1 is detailed in Sect. 5.2. However, determining the platform parameters α and φ_0 is challenging. Since entropy evaluation based on stochastic models essentially provides a lower bound of the entropy, only the worst-case scenario for α and φ_0 needs to be considered. According to [FBB+23], when $\alpha = \frac{K_D-1}{2K_D} \mod \frac{1}{K_D}$ and $\varphi_0 = \frac{\Delta}{2} \mod \Delta$, the average distance of the contributors reaches its maximum, minimizing the entropy contribution of these bits and representing the worst-case scenario. Without prejudice to generality, we select $\alpha = \frac{K_D - 1}{2K_D}$ and $\varphi_0 = \frac{\Delta}{2}$.

Table 3: Entropy evaluation parameter values

Category	Symbol	Value	Category	Symbol	Value
p_d	f_{in}	100MHz	p_p	α	$(K_D - 1)/K_D$
	N_{PLL}	2		φ_0	$\Delta/2$
	$f_{\text{-3dB}}$	4MHz		σ^2	11.79ps^2
	N_{output}	2			
	τ	90°			
	ς	{43,4,7,32,3,3}			

Based on the above analysis, the parameter values chosen for entropy rate assessment are shown in Table 3. According to the AIS-31 [KS11] standard, the entropy of the raw random numbers should be greater than 0.997. With $H_1 = 0.997$ and the evaluation parameters from Table 3, the required minimum jitter variance $\sigma^2_{\min}$ is calculated to be 9.24ps^2, which satisfies the condition $\sigma^2 > \sigma^2_{\min}$. The evaluation results indicate that even in the worst case, the measured jitter variance exceeds the threshold. Table 4 shows the results of the entropy evaluation. By taking into account the impact of PLL platform parameters, our stochastic model yields a more conservative lower bound for entropy compared to [FBB+23]. It is important to note that this result is achieved through careful control of the contributors as detailed in Sect. 5.1. For typical PLL configurations, the discrepancy in entropy estimation is expected to be even more significant.

Table 4: Results of the entropy evaluation

Type	Stoch.mod.		Stoch.mod. in [FBB+23]		Test suite	
	H_1	H_∞	H_1	H_∞	H_1(AIS-31)	H_∞(NIST)
Results	0.9983	0.9324	0.9992	0.9544	1.000	0.9374

Evaluation method comparison According to the TRNG design methodology, randomness assessment should be as close as possible to the entropy source. Therefore we base the entropy evaluation on the jitter of PLL. Table 5 compares our method with those used in previous studies.

Table 5: Entropy evaluation methods comparison

Method	This work	[FD02]	[FDŠB04]	[FBB19]	[FBB+23]
Entropy source	**1~2 PLL w/ multiple outputs**	1 PLL	1~2 PLL	1~2 PLL	1~2 PLL w/ multiple outputs
Stochastic model	✓	✗	✗	✓	✓
Jitter measurement	✓	✗	✓	✗	✗
PLL platform parameter	✓	✗	✗	✗	✗
Evaluation location	**Entropy source**	✗	Entropy source	Digitization module	Digitization module
Evaluation resolution	σ_{lt}/d_{step}[1]	✗	$\sigma_p/\max(\Delta T_{\min})$[2]	σ_p/Δ[3]	σ_p/Δ[3]
Jitter magnitude	3.43ps	✗	47-64ps[4]	8.72ps[5]	10.26ps[6]
Platform	**Artix 7 FPGA**	Altera FPLD	Altera FPLD	Cyclone III FPGA	Cyclone V FPGA Spartan 6 FPGA SmartFusion2 FPGA

[1] d_{step} is the average delay of the delay chain.
[2] $\max(\Delta T_{\min}) = T_0/(4K_D) \cdot \gcd(2K_M, K_D)$
[3] $\Delta = T_1/K_D$ is the resolution of the coherent sampling.
[4] from Sect. 2.3 in [FDŠB04].
[5] σ for Configuration #1 from Table 4 in [FBB19].
[6] $\sigma_{\min}$ from Fig. 7 in [FBB+23].

The PLL-TRNG described in [FD02] only verifies the statistical properties of the generated random numbers and does not perform an entropy rate assessment, which poses potential security risks.

[FDŠB04] provides a direct evaluation of the entropy source but the jitter is obtained by oscilloscope observations, which is not an on-chip method. The sampling process of the oscilloscope may include low-pass filtering effects, potentially leading to lower measurement accuracy. Furthermore, the evaluation criterion $\sigma > \max(\Delta T_{\min})$ is essentially a qualitative bias. Due to the lack of a stochastic model, designers cannot accurately perform entropy evaluation based on jitter.

[FBB19] and [FBB+23] both evaluate entropy after the digitalization module, with slight differences. [FBB19] requires the probability and variance of the contributors be greater than a certain threshold value, while [FBB+23] requires the counter variance to be greater than a low bound. Both methods do not directly evaluate the entropy source. Since PLLs typically operate at high frequencies, the metastability of flip-flops or TDCs can affect the measurement results. The measurement resolution depends on the relative amplitude between the period jitter σ_p and the coherent sampling resolution Δ. Typically, in order to obtain sufficient entropy, Δ is around 10ps or even smaller.

Correspondingly, the resolution of our evaluation method depends on the relative magnitude of the long-term jitter σ_{lt} and the average delay of DC d_{step}. As shown in Sect. 5.2.3, $d_{step} \approx 15$ns, which is similar to the value of Δ. However, the long-term jitter variance is significantly larger than the period jitter variance, and the relative magnitude depends on the ratio of the loop time constant $\tau_{loop} = 1/(2\pi f_{\text{-3dB}})$ to the clk_1 period T_1 [DDS18]. In our design, $\tau_{loop} \approx 0.04\mu$s and $T_1 \approx 2.8$ns yields that $\frac{\sigma_{lt}/d_{step}}{\sigma_p/\Delta} \approx 1.8$. This indicates that our evaluation method can achieve a 1.8 times improvement in jitter resolution. Thanks to the higher jitter resolution, our evaluation achieves the smallest jitter, although it is conducted on a different FPGA platform.

6.2 Operating condition fluctuations

We examine the entropy density (according to NIST SP 800-90B non-IID tests) and jitter measurements of PLL-TRNGs across different environmental temperatures and supply voltages. Experimental results reveal that, due to the negative feedback mechanism of PLLs, PLL-TRNGs demonstrate resilience to temperature and supply voltage fluctuations.

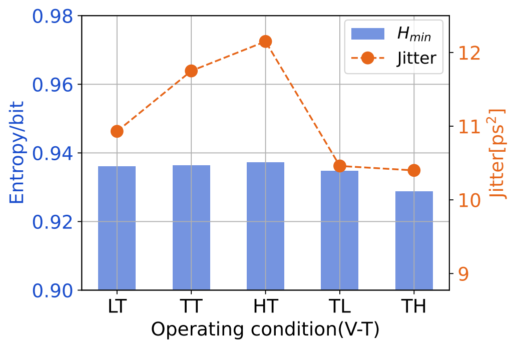


Figure 17: Entropy density and jitter measurements of PLL-TRNGs under different operating conditions. The x-axis indicates the operating conditions for supply voltages of 0.95V, 1.00V, and 1.05V (L-,T-,H-) and temperatures of 0°C, 25°C, and 85°C (-L,-T,-H).

6.3 Characterization of PLL phase noise

An ideal entropy source exhibits IID characteristics, corresponding to a spectrally flat phase noise profile across all frequencies. However, practical oscillators typically deviate from this ideal behavior. Unlike ideal entrpy sources, PLL phase noise exhibits stationarity and weak correlation, characterized by an exponentially decaying ACF. When accounting for thermal noise, the PLL maintains stationary output while approximating - though not fully achieving - output independence, thereby providing a practical approximation to an ideal entropy source. In our model, the effect of flicker noise on PLL phase noise is not considered. The presence of flicker noise inevitably destroys the stationarity of the PLL output. However, since the $1/f^3$ noise of the VCO is attenuated by the transfer function at low frequencies according to Eq. (8), it is still possible to approximate a stationary entropy source if we can ensure that the input flicker noise component is as small as possible. In summary, the stationarity of output and effective suppression of flicker noise constitute two advantages of PLLs as entropy sources.

6.4 Stochastic model in the case of ASIC platform

PLLs are easily available in mainstream FPGA platforms. As a result, PLL-TRNGs are predominantly applied in FPGA platforms. For ASIC-implemented PLLs, the phase noise PSD is similar to the derivation presented in Sect. 3.1 as shown in [Lee02]. Therefore, the stochastic model we propose is generic and applicable to both FPGA and ASIC implementations. Benefiting from the full-custom design flexibility, ASIC-implemented PLLs exhibit more adaptable frequency division ratios and bandwidths, typically operating beyond the GHz range. This capability significantly enhances the entropy rate of PLL-TRNGs. The design of PLL-TRNGs for ASIC platforms will be a focus of future research.

7 Conclusion

In this work, we extract key platform-specific parameters of the PLL, specifically tailored for PLL-TRNG design. Based on the Ornstein-Uhlenbeck process, we propose a new stochastic model for multiple PLL outputs. Our stochastic model simplifies the verification of the model assumptions in [FBB+23].

We evaluate the effect of different parameters on correlation and entropy using Monte Carlo simulation. We conclude that the intra-mode bit correlation is lower than the phase correlation, allowing us to quantitatively estimate the minimum time lag required for a given correlation threshold. To the best of our knowledge, we are the first to reveal the effect of the PLL's bandwidth on the correlation of sampling points.

We propose an entropy evaluation method incorporating jitter measurements into the stochastic model. Experimental results demonstrate that the model accurately characterizes the PLL's jitter. Additionally, we show that the entropy evaluation method achieves approximately a 1.8-fold improvement in jitter resolution. Based on this, we can obtain the most conservative entropy estimation. Our work focuses on design methods for entropy sources, with plans for future work including online tests, total failure tests, and potential post-processing.

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References

[BBF+18] Josep Balasch, Florent Bernard, Viktor Fischer, Miloš Grujić, Marek Laban, Oto Petura, Vladimir Rožić, Gerard van Battum, Ingrid Verbauwhede, Marnix Wakker, and Bohan Yang. Design and testing methodologies for true random number generators towards industry certification. In *2018 IEEE 23rd European Test Symposium (ETS)*, pages 1–10, May 2018.

[BCPW22] L. Benea, M. Carmona, F. Pebay-Peyroula, and R. Wacquez. On the Characterization of Jitter in Ring Oscillators using Allan variance for True Random Number Generator Applications. In *2022 25th Euromicro Conference on Digital System Design (DSD)*, pages 534–538, August 2022.

[BFV10] Florent Bernard, Viktor Fischer, and Boyan Valtchanov. Mathematical Model of Physical RNGs Based On Coherent Sampling. *Tatra Mountains - Mathematical Publications*, 45, August 2010.

[BGH+24] Florent Bernard, Arturo Garay, Patrick Haddad, Nathalie Bochard, and Viktor Fischer. Low Cost and Precise Jitter Measurement Method for TRNG Entropy Assessment. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2024(1):207–228, 2024.

[BLMT11] Mathieu Baudet, David Lubicz, Julien Micolod, and André Tassiaux. On the Security of Oscillator-Based Random Number Generators. *Journal of Cryptology*, 24(2):398–425, April 2011.

[Cic03] Cicero S. Vaucher, Bram Nauta. *Architectures for RF Frequency Synthesizers*, volume 693 of *The International Series in Engineering and Computer Science.* Kluwer Academic Publishers, Boston, 2003.

[DDS18] Nicola Da Dalt and Ali Sheikholeslami. *Understanding Jitter and Phase Noise: A Circuits and Systems Perspective.* Cambridge University Press, 1 edition, February 2018.

[FBB19] Viktor Fischer, Florent Bernard, and Nathalie Bochard. Modern random number generator design – Case study on a secured PLL-based TRNG. *it - Information Technology*, 61(1):3–13, February 2019.

[FBB+23] Viktor Fischer, Florent Bernard, Nathalie Bochard, Quentin Dallison, and Maciej Skorski. Enhancing Quality and Security of the PLL-TRNG. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2023(4):211–237, August 2023.

[FD02] Viktor Fischer and Miloš Drutarovský. True Random Number Generator Embedded in Reconfigurable Hardware. In *Cryptographic Hardware and Embedded Systems*, volume 2523, pages 415–430, August 2002.

[FDŠB04] Viktor Fischer, Miloš Drutarovský, Martin Šimka, and Nathalie Bochard. High Performance True Random Number Generator in Altera Stratix FPLDs. In *Field Programmable Logic and Application*, volume 3203, pages 555–564, Berlin, Heidelberg, 2004. Springer Berlin Heidelberg.

[GGFZ22] Davide Galli, Andrea Galimberti, William Fornaciari, and Davide Zoni. On the Effectiveness of True Random Number Generators Implemented on FPGAs. In Alex Orailoglu, Marc Reichenbach, and Matthias Jung, editors, *Embedded Computer Systems: Architectures, Modeling, and Simulation*, volume 13511, pages 315–326. Springer International Publishing, Cham, 2022.

[HTBF14] Patrick Haddad, Yannick Teglia, Florent Bernard, and Viktor Fischer. On the assumption of mutual independence of jitter realizations in P-TRNG stochastic models. In *2014 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 1–6, March 2014.

[KS11] W. Killmann and W. Schindler. Ais20/ais31. a proposal for: Functionality classes for random number generators, version 2.0, 2011. [Online].

[LBGD13] Florent Lozach, Molka Ben-Romdhane, Tarik Graba, and Jean-Luc Danger. FPGA Design of an Open-Loop True Random Number Generator. In *2013 Euromicro Conference on Digital System Design*, pages 615–622, September 2013.

[Lee02] D.C. Lee. Analysis of jitter in phase-locked loops. *IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing*, 49(11):704–711, November 2002.

[Meh02] A. Mehrotra. Noise analysis of phase-locked loops. *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, 49(9):1309–1316, September 2002.

[MGBF+23] Arturo Mollinedo Garay, Florent Bernard, Viktor Fischer, Patrick Haddad, and Ugo Mureddu. An Evaluation Procedure for Comparing Clock Jitter Measurement Methods. In Ileana Buhan and Tobias Schneider, editors, *Smart Card Research and Advanced Applications*, Lecture Notes in Computer Science, pages 167–187, Cham, 2023. Springer International Publishing.

[PMB+16] Oto Petura, Ugo Mureddu, Nathalie Bochard, Viktor Fischer, and Lilian Bossuet. A survey of AIS-20/31 compliant TRNG cores suitable for FPGA devices. In *2016 26th International Conference on Field Programmable Logic and Applications (FPL)*, pages 1–10, Lausanne, Switzerland, August 2016. IEEE.

[PMBF17] Oto Petura, Ugo Mureddu, Nathalie Bochard, and Viktor Fischer. Optimization of the PLL based TRNG design using the genetic algorithm. In *2017 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–4, Baltimore, MD, USA, May 2017. IEEE.

[PV22] Adriaan Peetermans and Ingrid Verbauwhede. An energy and area efficient, all digital entropy source compatible with modern standards based on jitter pipelining. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 88–109, August 2022.

[PV24a] Adriaan Peetermans and Ingrid Verbauwhede. Characterization of Oscillator Phase Noise Arising From Multiple Sources for ASIC True Random Number Generation. *IEEE Transactions on Circuits and Systems I: Regular Papers*, pages 1–14, 2024.

[PV24b] Adriaan Peetermans and Ingrid Verbauwhede. TRNG Entropy Model in the Presence of Flicker FM Noise. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2024, 2024.

[SDFF06] M. Simka, M. Drutarovsky, V. Fischer, and J. Fayolle. Model of a true random number generator aimed at cryptographic applications. In *2006 IEEE International Symposium on Circuits and Systems*, page 4, Island of Kos, Greece, 2006. IEEE.

[SMS07] Berk Sunar, William J. Martin, and Douglas R. Stinson. A Provably Secure True Random Number Generator with Built-In Tolerance to Active Attacks. *IEEE Transactions on Computers*, 56(1):109–119, January 2007.

[TBK+18] M. Turan, E. Barker, J. Kelsey, K. McKay, M. Baish, and M. Boyle. Recommendation for the entropy sources used for random bit generation. Technical Report NIST Special Publication 800-90B, National Institute of Standards and Technology (NIST), 2018. [Online].

[VFAB10] Boyan Valtchanov, Viktor Fischer, Alain Aubert, and Florent Bernard. Characterization of randomness sources in ring oscillator-based true random number generators in FPGAs. In *13th IEEE Symposium on Design and Diagnostics of Electronic Circuits and Systems*, pages 48–53, April 2010.

[XKGN09] Xiang Gao, E.A.M. Klumperink, P.F.J. Geraedts, and B. Nauta. Jitter Analysis and a Benchmarking Figure-of-Merit for Phase-Locked Loops. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 56(2):117–121, February 2009.

[YRG+17] Bohan Yang, Vladimir Rožić, Miloš Grujić, Nele Mentens, and Ingrid Verbauwhede. On-chip jitter measurement for true random number generators. In *2017 Asian Hardware Oriented Security and Trust Symposium (AsianHOST)*, pages 91–96, Beijing, China, October 2017. IEEE.